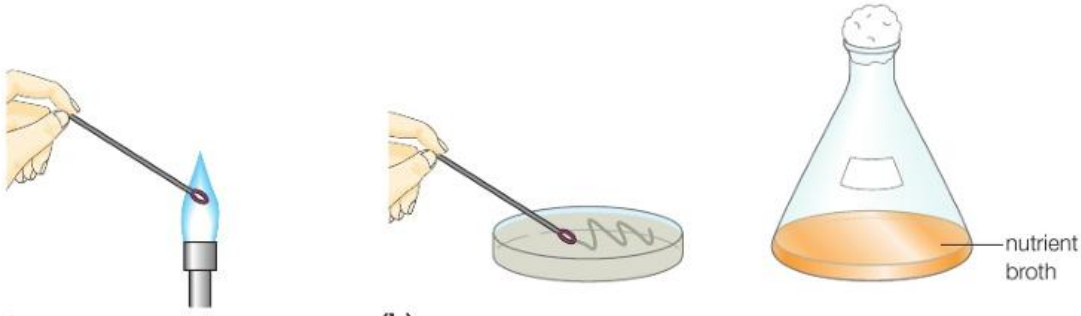


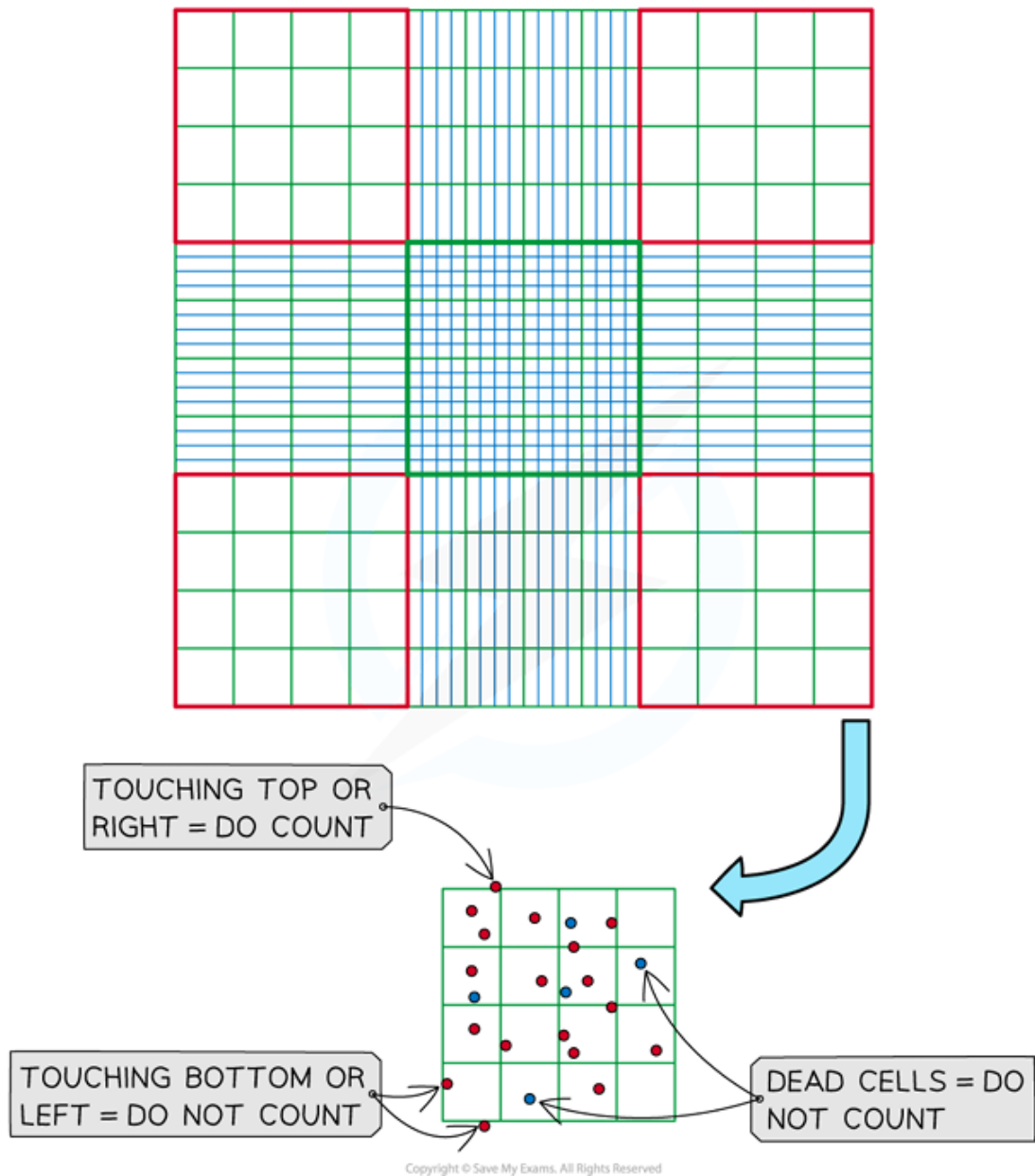
Topic 6 – Microbiology, Immunity and Forensics

Candidates will be assessed on their ability to:

6.1	understand the principles and techniques involved in culturing microorganisms, using aseptic technique
	 (a) Sterilise the inoculating loop by holding it in the Bunsen burner until it glows red hot and then leave it to cool. (b) Dip the sterilised loop in bacteria on the surface of an agar plate or any other chosen sample. Then swirl the loop in the sterilised liquid medium in a conical flask. Remove the loop and seal the flask with sterile cotton wool. Label clearly before incubating at a suitable temperature. ▲ fig C Inoculating a flask with bacteria. Aseptic Techniques: methods used to prevent contamination of culture with other organisms 1. Carry out the work beside a Bunsen burner to move microorganisms in the air away 2. Use sterilized equipment to kill any bacteria present 3. Wash hands and wear gloves to prevent transferring bacteria into cultures 4. Minimize the time that cultures are exposed to the air to reduce change of contamination 5. Close windows to avoid draughts Why Aseptic Technique is important: - To prevent entry of bacteria that may compete with culturing organism for nutrient or - To prevent the entry of bacteria that may produce toxins to the bacteria being cultured - To prevent infection of person with bacteria in the culture that maybe pathogenic - To prevent entry of bacteria that may grow in different concentration of microbial substances Culturing Microorganisms: - Use aseptic techniques - Identify source of sample - Use selective media / antibiotics / color of colonies to identify bacteria - Culture the microorganism on agar / in a broth in nutrients - Incubate at a temperature between 25°C to 35°C: optimum for enzymes - Appropriate pH condition: use buffer - Appropriate length to time in the range of 24 hours to 1 week: to ensure bacterial growth to be visible How Petri dishes should be sealed while testing effectiveness of antibiotics: - It is held on with tape but not completely sealed - It must be sealed to prevent the lid from falling off and contamination occurring and to prevent growth of anaerobic bacteria

6.2

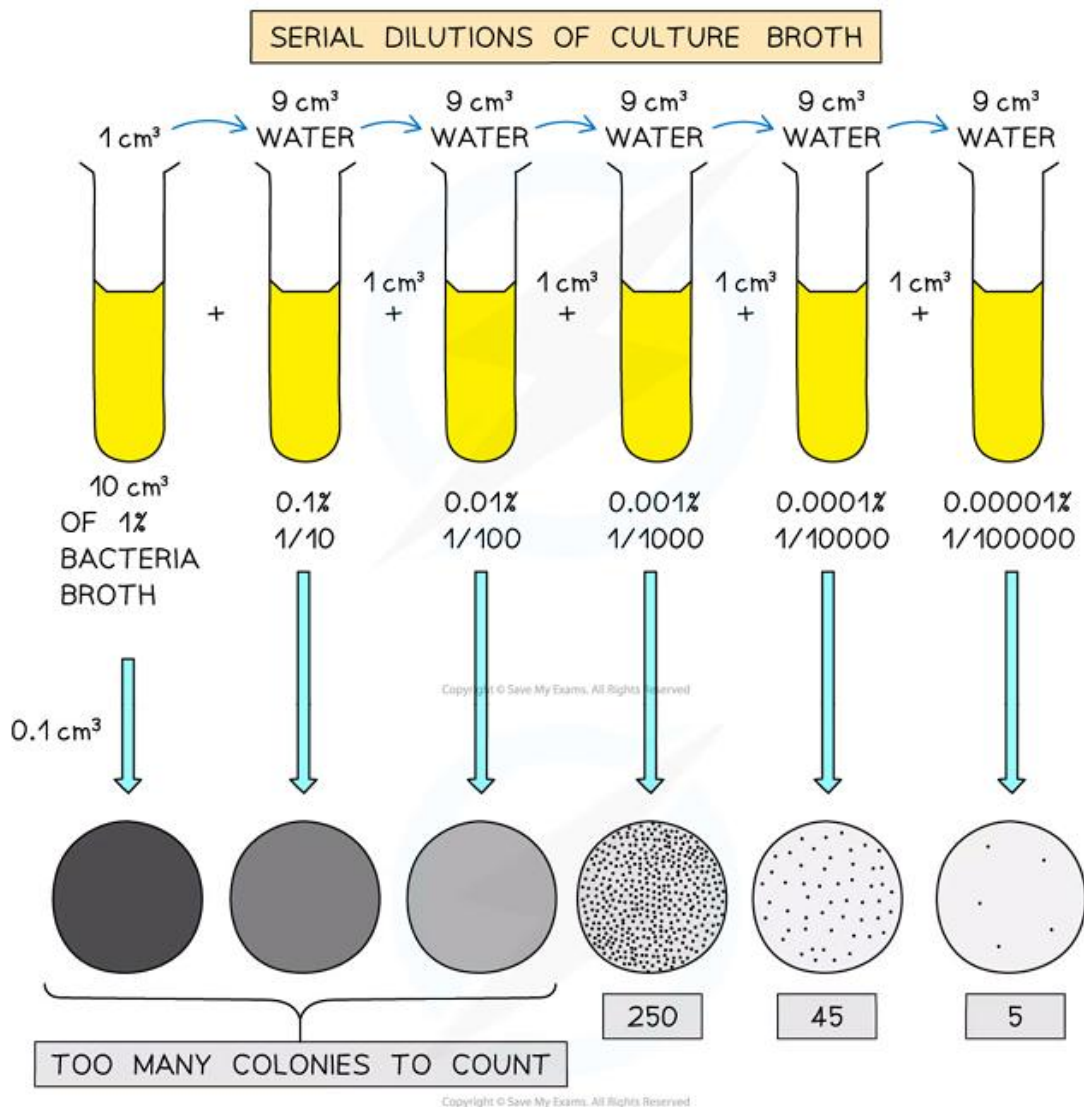
understand the different methods of measuring the growth of microorganisms, as illustrated by cell counts, dilution plating, mass and optical methods (turbidity)



Haemocytometer: a specialized thick microscope slide with a rectangular chamber that holds a standard volume of liquid of 0.1 mm^3

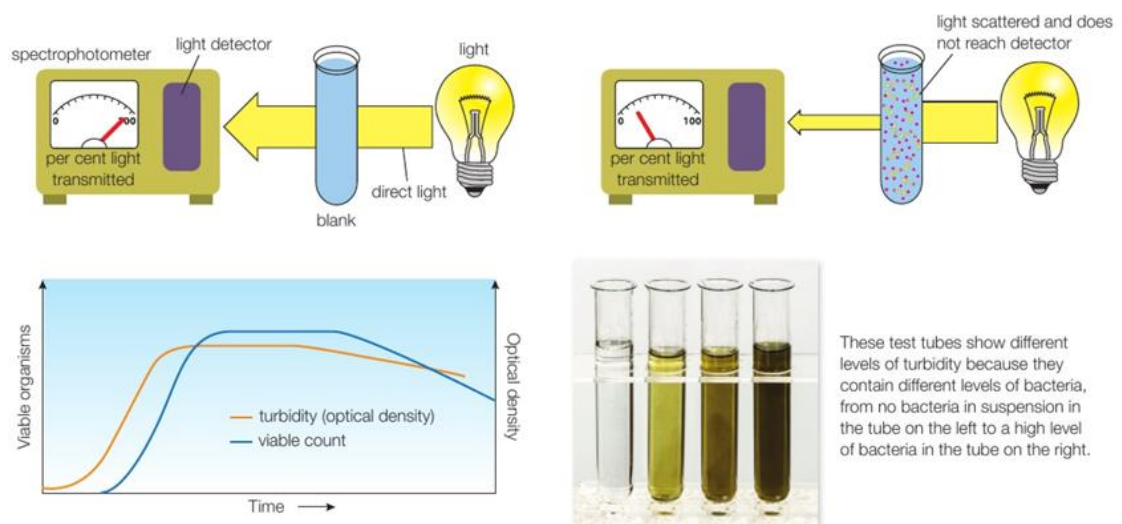
- Dilute sample of nutrient broth by half with an equal volume of trypan blue (a dye that stains dead cells blue)
- View and count the cells using a microscope

No. of cells per ml nutrient broth = mean cell count x dilution factor x 10^4



Dilution Plating:

- Use serial dilutions to reduce the number of bacteria in the samples to be able to count
- Make a bacterial lawn on agar to separate individual bacteria
- Keep for a fixed time and temperature to be able to count colonies
- Dilution Plating only measure living bacteria count



Optical Method (Turbidity):

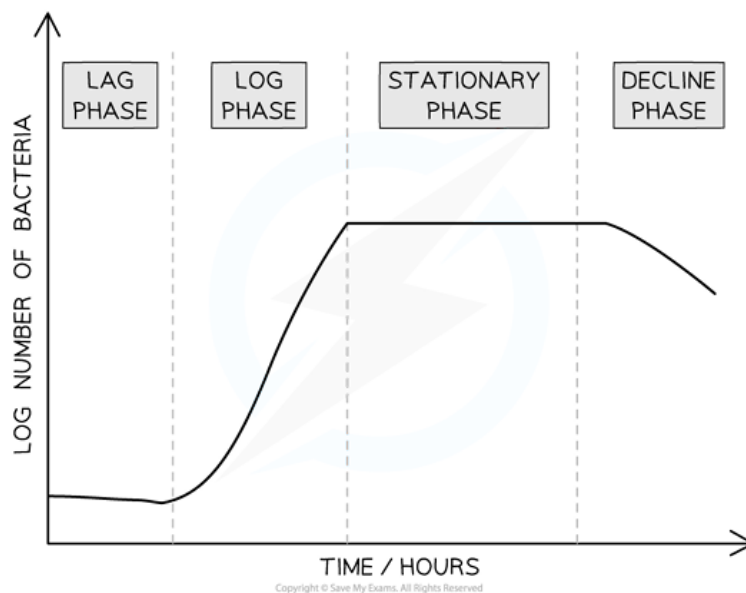
- Use colorimeter
- Calibration curve used to determine concentration of bacteria
→ This should be done with other methods such as haemocytometer because turbidity alone cannot determine concentration
→ It can only give you relative abundance
- Optical method measures total cell counts (both dead and alive cells)

Mass and Area:

- Fungi: measure diameter of individual areas of mycelium to determine growth
→ Dry mass: Removes samples of broth at regular intervals and separate the fungi from the liquid by centrifugation or filtering; then dry the material and find the dry mass

6.3

understand the different phases of a bacterial growth curve (lag phase, exponential phase, stationary phase and death phase) and be able to calculate exponential growth rate constants

**Bacterial Growth Curve:**

1. **Lag** Phase: No increase in number of bacteria because the cells had to adjust to new environment
2. **Exponential** Phase: Increase in number of bacteria as conditions are suitable (nutrients available / pH appropriate / oxygen levels appropriate / appropriate temperature)
3. **Stationary** Phase: Rate of growth is equal to rate of death
4. **Death** Phase: Rate of death is greater than rate of growth

Calculating Exponential Growth Rate:

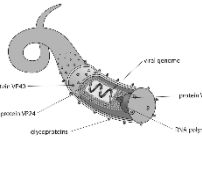
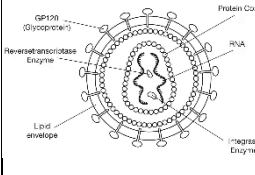
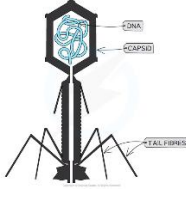
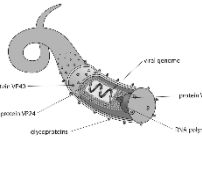
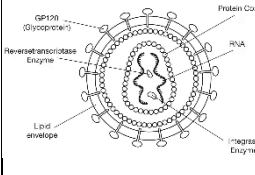
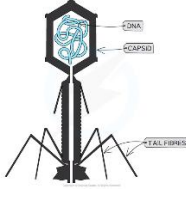
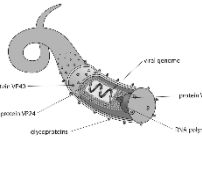
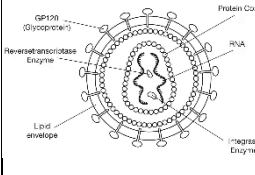
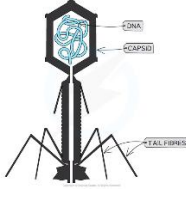
To find number of organisms at time t:

$$N_t = N_0 \times 2^{kt}$$

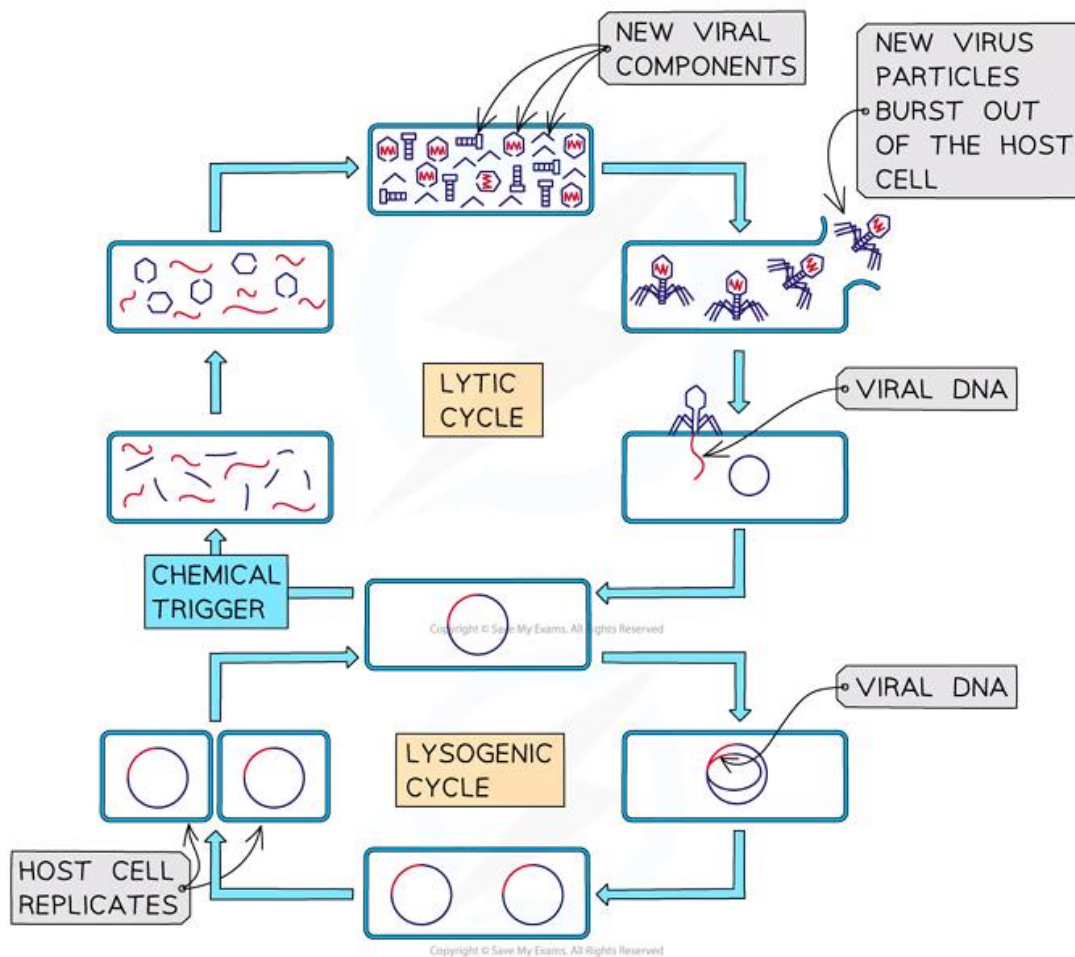
To find exponential growth

rate constant k:

$$k = \frac{\log_{10} N_t - \log_{10} N_0}{\log_{10} 2 \times t}$$

	N_t = the number of organisms at time t N_0 = the number of organisms at time 0 (at the start) k = the exponential growth rate constant t = the time for which the colony has been growing																														
6.4	CORE PRACTIAL 13 Investigate the rate of growth of microorganisms in a liquid culture, taking into account the safe and ethical use of organisms.																														
6.5	(i) be able to compare the structure of bacteria and viruses (nucleic acid, capsid structure and envelope) with reference to Ebola virus, tobacco mosaic virus (TMV), human immunodeficiency virus (HIV) and lambda phage (ii) understand what is meant by the terms <i>lytic</i> and <i>latency</i>																														
i	<table><tr><th></th><th>Ebola Virus</th><th>Tobacco Mosaic Virus (TMV)</th><th>Human Immunodeficiency Virus (HIV)</th><th>Lambda Phage (λ phage)</th></tr><tr><td>Lipid Envelope</td><td>✓</td><td>✗</td><td>✓</td><td>✗</td></tr><tr><td>Capsid Structure</td><td>Helical Elongated Shape</td><td>Helical Polypeptide</td><td>Polyhedral</td><td>Icosahedral Head; Complex</td></tr><tr><td>Genetic Material</td><td>Single Stranded RNA</td><td>Single Stranded RNA</td><td>(2) RNA Molecules</td><td>DNA molecules; Double Stranded</td></tr><tr><td>Host</td><td>Infect endothelial cells and liver cells</td><td>Infect T helper cells</td><td>Infect plants</td><td>Infect bacteria</td></tr><tr><td>Diagram</td><td></td><td></td><td></td><td></td></tr></table> Other Commonly Asked Viruses: - Coronavirus: Helical Capsid, Enveloped, RNA virus - Adenovirus: Icosahedral, Non-enveloped, DNA virus - Influenza: Helical Capsid, Enveloped, RNA virus - Rabies: Complex, Enveloped, RNA virus		Ebola Virus	Tobacco Mosaic Virus (TMV)	Human Immunodeficiency Virus (HIV)	Lambda Phage (λ phage)	Lipid Envelope	✓	✗	✓	✗	Capsid Structure	Helical Elongated Shape	Helical Polypeptide	Polyhedral	Icosahedral Head; Complex	Genetic Material	Single Stranded RNA	Single Stranded RNA	(2) RNA Molecules	DNA molecules; Double Stranded	Host	Infect endothelial cells and liver cells	Infect T helper cells	Infect plants	Infect bacteria	Diagram				
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Host	Infect endothelial cells and liver cells	Infect T helper cells	Infect plants	Infect bacteria																											
Diagram																															

ii



Lysogenic Pathway:

- Viral DNA inserted into the host DNA to form a provirus
- mRNA is not produced from viral DNA yet due to a repressor protein
- During this period of lysogeny, virus is latent
- Under certain condition like malnutrition, the virus enter lytic pathway.

Lytic Pathway:

- Viral genetic material is replicated independently of the host DNA immediately after it enters the host
- Mature viruses are made and eventually the host cell bursts

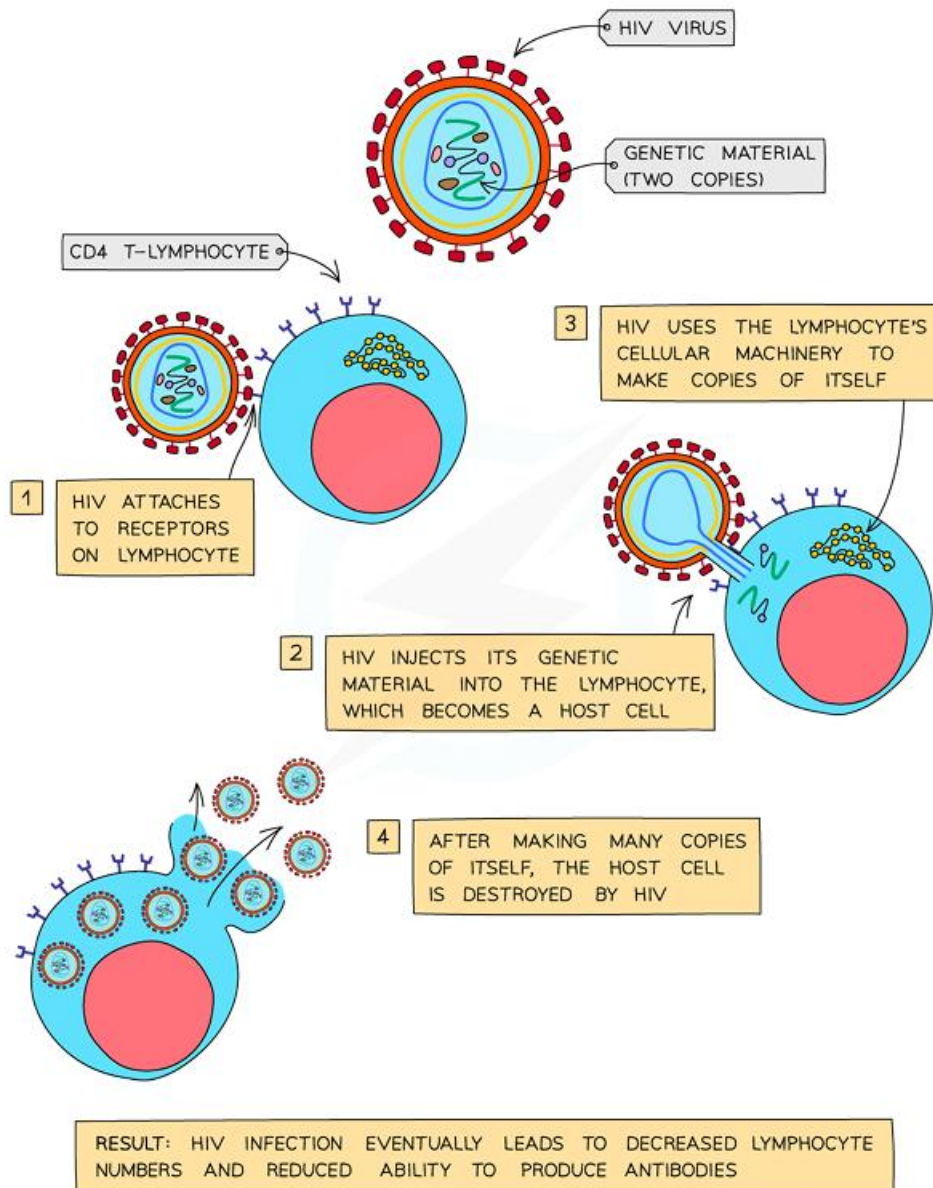
6.6

understand how *Mycobacterium tuberculosis* and human immunodeficiency virus (HIV) infect human cells, causing symptoms that may result in death

Mycobacterium Tuberculosis

How infection with *M. tuberculosis* may result in death:

- Formation of tubercles in lungs
- Reduce surface area of alveoli so reduce gas exchange
- Therefore, insufficient oxygen for aerobic respiration
- Causes other organ failure
- Weakens immune system so death from other infections



HIV

(2) Types of Cells HIV can infect:

1. T Helper Cells
2. Macrophages

How HIV is able to enter the cells:

- Gp120 protein (or Glycoprotein) on HIV binds to CD4 receptor on T helper cell or macrophage
- Fusion of virus envelope with cell surface membrane
- HIV can also enter into macrophage via phagocytosis

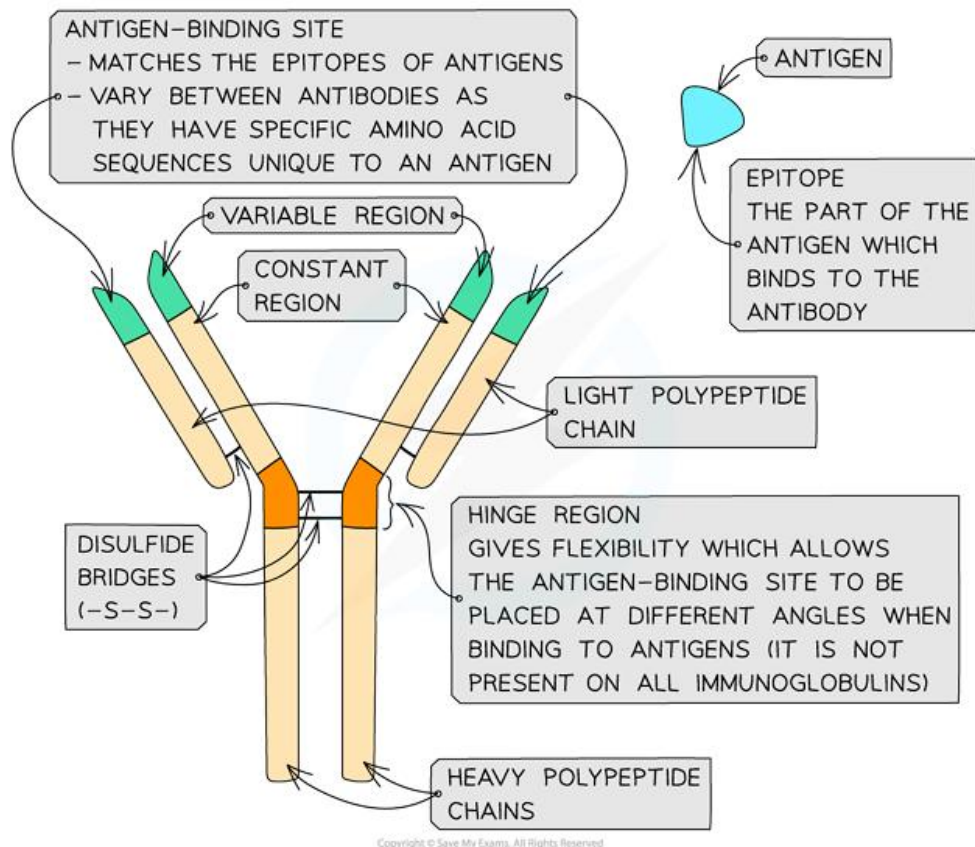
Infection of Human Immunodeficiency Virus:

1. Reverse transcriptase enzyme is used to produce viral DNA from viral RNA
2. Integrase enzyme is used to incorporate viral DNA into host cell's DNA to form a provirus
3. Viral RNA and proteins are produced and assembled to form viruses
4. These viruses infect further T helper cells
5. T killer cells destroy infected T helper cells
6. This lowers immunity because T helper cells produce cytokines, activate B cells and T killer cells
7. Death is caused by opportunistic disease such as TB, Kaposi's sarcoma and cancer

6.7	(i) know the major routes pathogens may take when entering the body (ii) understand the role of barriers in protecting the body from infection, including skin, stomach acid, and gut and skin flora
i	Methods of Infection: - Vectors: living organisms that transmits infections from one host to another - Inhalation: droplets expelled from respiratory tract; small droplets fall of pathogens in the air - Ingestion: caused by faecally contaminated food or drink - Fomites: inanimate objects that carry pathogens from one host to another - Direct Contact: from direct skin contact via genital organs or skin touch - Inoculation: inoculated in the body directly through a break in skin
ii	Infection: pathogen being inside the cells Barriers to Entry: 1. Skin: - Keratin: Fibrous structural protein and acts as physical barrier - Antimicrobial secretions by the skin like oil and sebum 2. Stomach Acid: - Stomach contains hydrochloric acid - pH is very low so most bacteria are destroyed 3. Gut and Skin Flora: - Prevent growth of the pathogen - This is because they compete with pathogens for space, nutrients and attachment sites to gut wall - Also, they release toxins to destroy pathogens
6.8	understand the non-specific responses of the body to infection, including inflammation, lysozyme action, interferon and phagocytosis
	Non-Specific Responses: 1. Inflammation - Swelling: More white blood cells and platelets to the wound - Redness: More blood and phagocytes and antibodies to the wound - Hot: Increase rate of enzyme activity of the person so speed up phagocytosis → Also reduce activity of bacterial enzymes so reduce growth of bacteria - Pain: To avoid contact with area and alerts the person that there is an injury 2. Lysozyme Action - Mucus in respiratory epithelium cells contains lysozymes which destroy bacterial cell wall 3. Interferon - Interferons are secreted by infected cells and they are anti-viral proteins - Inhibit viral replication - Prevents attachment of virus to other host cells so that other cells cannot be infected once the viruses burst out of the infected cell - Also involved in activation of macrophages 4. Phagocytosis - To engulf bacteria / virus / pathogens enclosed in phagosome - Phagosome fuses with digestive enzymes (lysozyme/protease) - Lysozyme breaks down the pathogen - After phagocytosis, macrophage can present antigen

6.9

understand the roles of antigens and antibodies in the body's immune response including the involvement of plasma cells, macrophages and antigen-presenting cells



Antibody Structure:

- Antibody is a glycoprotein
- Disulfide bridge, L and H region, Hinge region
- Antibody has constant region
- Antibody is produced by plasma cells

Functions of Antibody:

- Opsonization: Binds to the antigens on the foreign body for faster destruction by phagocytosis
- Agglutination / Immobilization: Cause pathogens to clump together for easy phagocytosis
- Toxin Neutralization: Bind to toxins produced by pathogen and neutralizes them

Antigen: substances that stimulates immune response

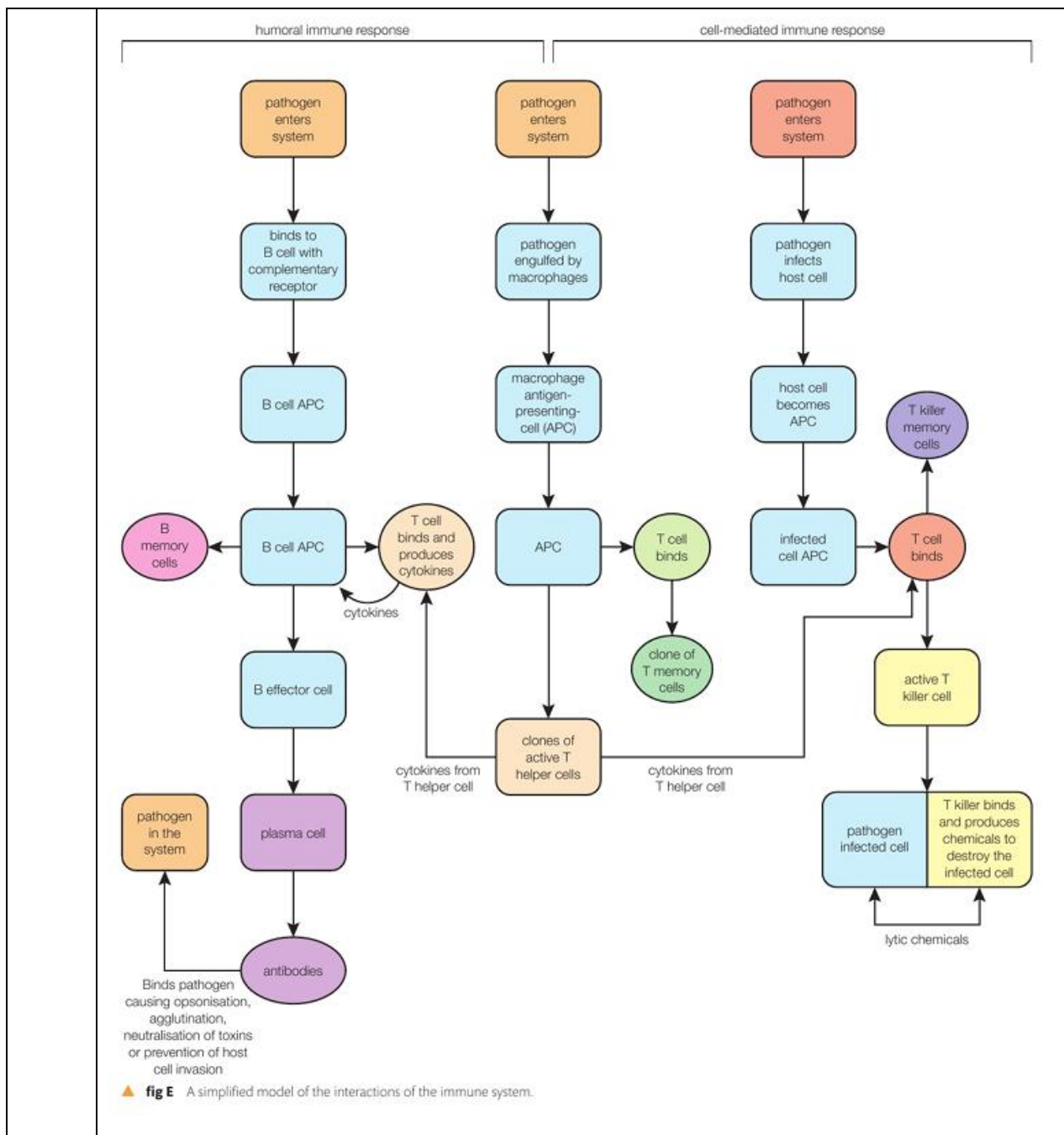
Antibody and Antigen:

- Antibody can only bind their specific antigen
- This is because of the complementary shape of antibody and antigen
- One antigen can bind to two bacteria (Variable Region) → Agglutination
- Antibody can bind to phagocyte (Constant Region) → Opsonization
- So, phagocytosis is enhanced

Macrophages:

- Microorganisms engulfed by the macrophage
- Microorganism is digested by the macrophage with enzymes
- Antigen of destroyed microorganism attach to MHC molecules on the surface of macrophage
- This is to represent the antigen to T helper cells
- Macrophage membrane can attach to antibody for opsonization

6.10	understand the differences between the roles of B cells (B memory and B effector cells), and T cells (T helper, T killer and T memory cells) in the host's immune response
	Immune Response: <pre> graph TD A[T helper Activation] --> B[Effector Stage] A --> C[Cell-mediated Response] </pre> T helper Activation: - Microorganisms engulfed by the macrophage - Microorganism is digested by the macrophage with enzymes - Antigen of destroyed microorganism attach to MHC molecules on the surface of macrophage - This is to represent the antigen to T helper cells - T helper cells become activated - T helper cells release cytokines to stimulate humoral response and cell-mediated response Effector Stage: - T helper releases cytokines that stimulate B cell division - B cells become plasma cells - So that antibody can be produced from plasma cells for opsonization, agglutination and antitoxin - B cells also generates B memory cells - This is for long term immunity (secondary immune response) Cell-mediated Response - T killer cell binds to infected cell (APC) - T killer cells destroy virus-infected cells by releasing chemicals - Macrophages destroy the virus particles - Memory cells made to protect from future infections



6.11 understand how individuals may develop immunity (natural, artificial, active and passive)

Natural Immunity:

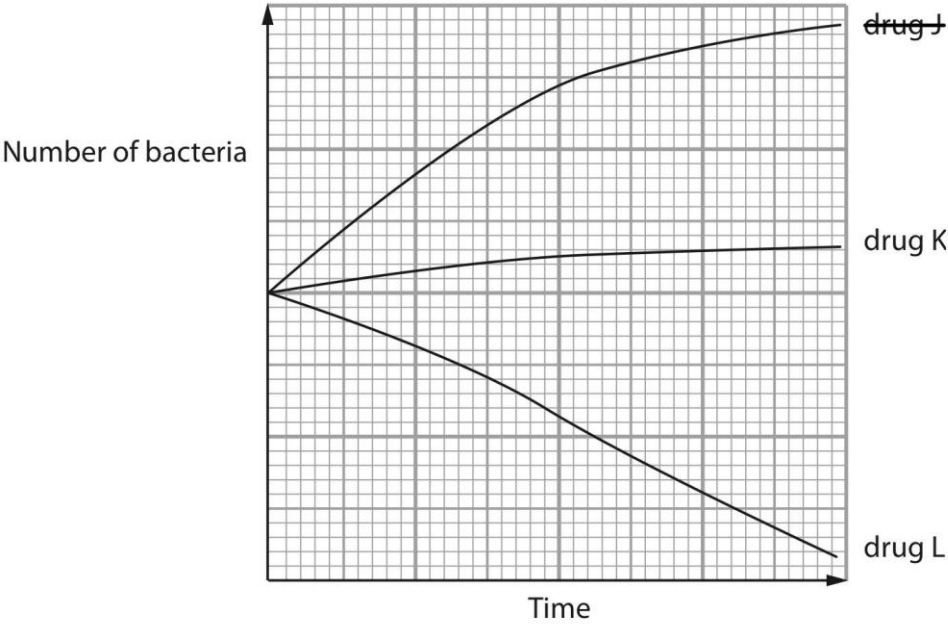
1. **Active:** When the body produces its own antibodies to an antigen encountered naturally
2. **Passive:** When antibodies made by the mother are passed to the baby via the placenta or mother's milk

Artificial Immunity:

1. **Active:** Inject attenuated pathogens to stimulate primary immune response to produce memory cells
2. **Passive:** When antibodies are extracted from one individual to another (tetanus vaccine)

How vaccine produce artificial active immunity:

- Inject the attenuated antigen
- Antigen engulfed by macrophage
- Macrophage presents antigen to T helper cells
- Formation of B and T memory cells

	→ Vaccination protect people who cannot become immune → Vaccinated people will not develop the disease → Herd immunity is attained
6.12	understand how the theory of an 'evolutionary race' between pathogens and their hosts is supported by evasion mechanisms shown by pathogens
	Antibiotic Resistance: - Antibiotics acts as selection pressure - When the antibiotic is being taken, the resistant bacteria will survive - When the antibiotic is not being taken the non-resistant bacteria will also survive - So, there will be more competition with non-resistant bacteria for nutrient and space so resistant bacteria decreases Evolutionary Race because: - The number of resistant bacteria is increasing - New antibiotics are being developed
6.13	understand the difference between bacteriostatic and bactericidal antibiotics
	 Bacteriostatic: a chemical that inhibits the growth of bacteria (drug K) Bactericidal: a chemical that kills bacteria (drug L)
6.14	CORE PRACTICAL 14 Investigate the effect of different antibiotics on bacteria.
6.15	know how an understanding of the contributory causes of hospital-acquired infections has led to codes of practice regarding antibiotic prescription and hospital practice that relate to infection prevention and control
	Hospital Acquired Infections (HAIs): infections that are acquired by a patient while in hospital Causes: - Poor hygiene practices - Weakened immune systems of patient - Antibiotic resistant bacteria: MRSA and Clostridium difficile

Codes of Practice:**- Doctor's Antibiotic Prescription:**

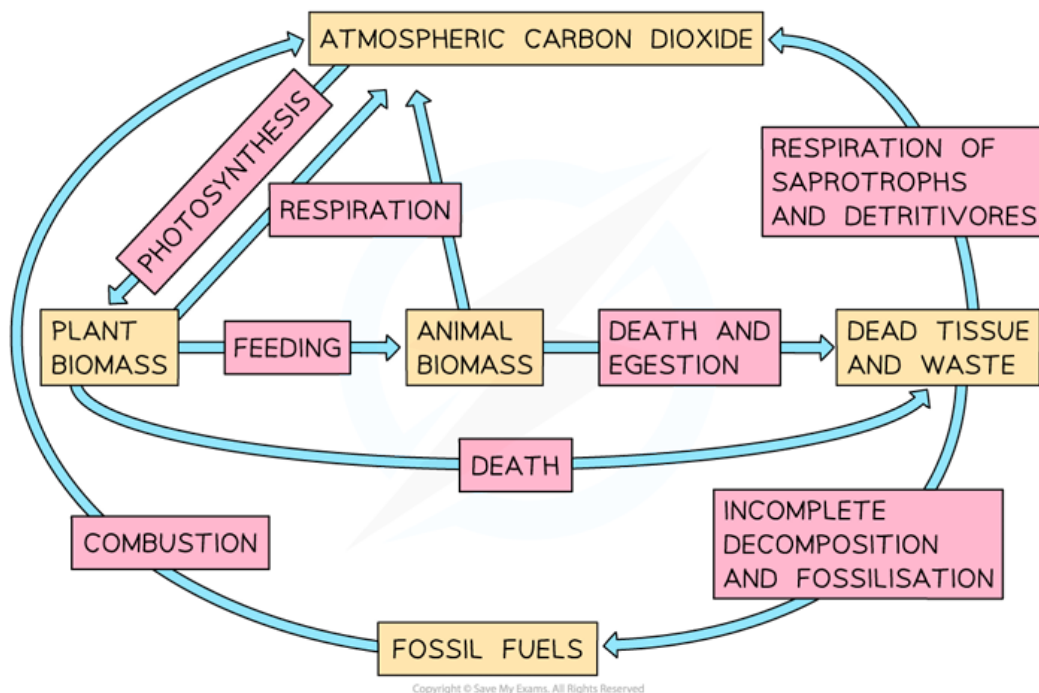
- Finish the course of antibiotics
- Take the antibiotics as directed by the doctor
- Warning and precautions should be given to the patient
- Use narrow spectrum antibiotic for the infection

- Hospital Practice:

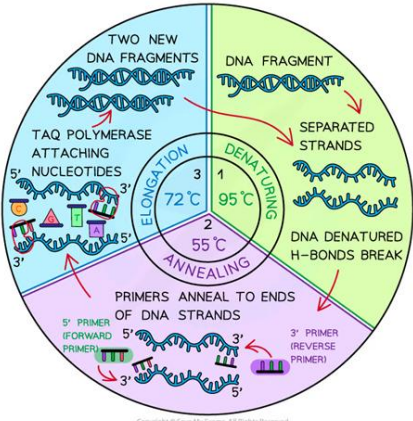
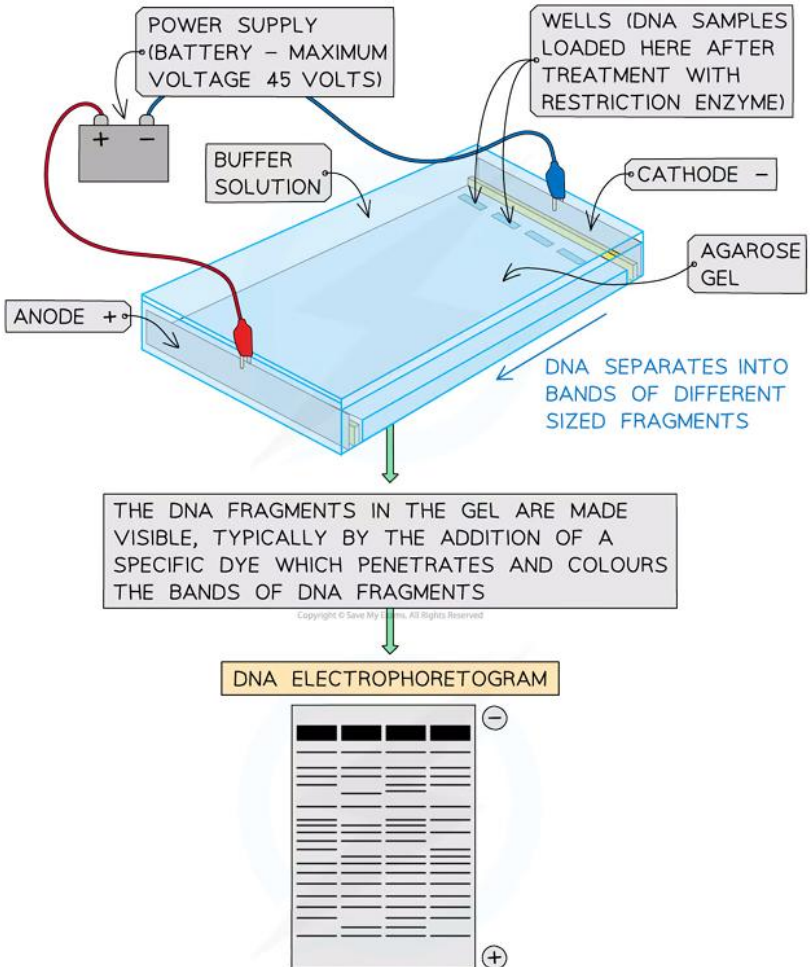
- Hand washing
- Washing bedding frequently
- Screening patients and isolate infected patient
- Removal of outdoor clothing
- Wearing protective clothing

6.16

know the role of microorganisms in the decomposition of organic matter and the recycling of carbon

**How decomposers involve in carbon cycle:**

- Bacteria and fungi are decomposers
- Decomposers break down organic matters
- Decomposers release enzymes for decomposition
- Digested molecules are absorbed into the decomposer
- Aerobic respiration release carbon dioxide
- Break down carbohydrates using water to hydrolyze the glyosidic bonds
- Monosaccharides are taken up by diffusion

6.17	know how DNA can be amplified using the polymerase chain reaction (PCR)
	Molecules Required for PCR: - DNA source - Taq/DNA Polymerase - Primers - DNA mononucleotides - Buffer Process of PCR: 1. Denaturation: - 94°C to break hydrogen bonds between the strands 2. Annealing: - 54°C to allow primers to attach - Two primers are needed to amplify one molecule of DNA → Primers are needed so that DNA polymerase can bind to one stand → One primer is needed for each strand of DNA → DNA stands have different base sequences at each end 3. Elongation: - 72°C for new mononucleotides can join to base to form two molecules - This is optimum temperature for Taq DNA polymerase enzyme to build complementary stands of DNA 4. Many cycles are run so that several molecules of DNA are made 
6.18	know how gel electrophoresis can be used to separate DNA fragments of different length
	

Process of Gel Electrophoresis (DNA):

1. Multiple copies of DNA made using PCR
2. Restriction enzyme cut the DNA into fragments
3. Load the DNA onto the agarose gel
4. Electric current is applied
5. Fluorescent tag is used to visualize the bands

Process of Gel Electrophoresis (Protein):

1. Extract protein from blood
2. Cut the protein into fragments using protease enzymes to hydrolyze peptide bonds
3. Load onto the gel
4. Electric field applied to the gel
5. Peptides are stained
6. Compare the peptide bands' width, size and pattern

Explain why restriction enzyme cuts the DNA into fragments of different lengths that can be separated by gel electrophoresis:

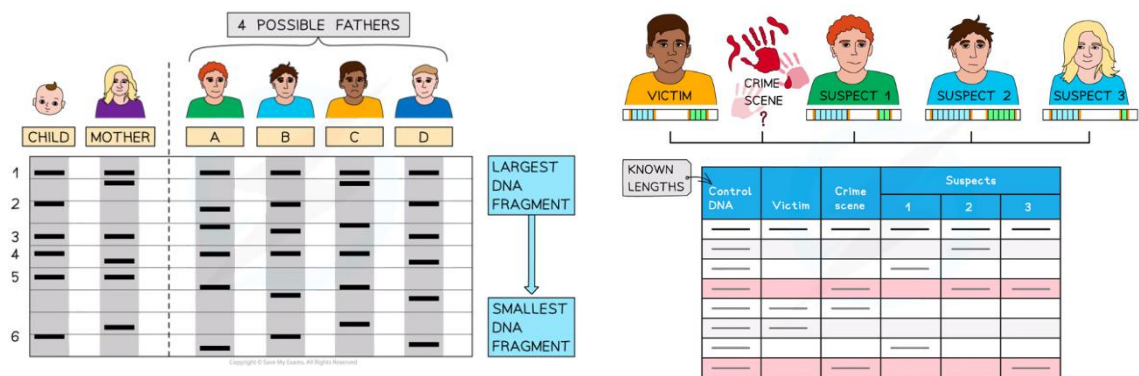
- Restriction enzyme recognize specific base sequences
- These recognition sites are not equally spaced along the DNA
- The enzyme hydrolyses the phosphodiester bonds
- Therefore, different sized fragments move different distances through the gel

Analysis of Bands:

- Thickest Band → Most Abundant Fragment
- Furthest Band → Most Soluble Fragment in the Solvent Used / Lightest Mass Fragment

6.19

understand how DNA profiling is used for identification and determining genetic relationships between organisms (plants and animals)

**DNA Profiling**

- A DNA profile can be produced using the process of gel electrophoresis
- The result is a series of bands of DNA of different length which can be seen due to the addition of radioactive or fluorescent labels

How to make a DNA Profile:

1. Collect sample of the organism (blood, nail, hair)
2. Extract DNA and amplify it using PCR
3. Use restriction enzyme to cut the DNA
4. DNA fragments are separated by size in the gel
5. Bands are stained to make them visible to produce fingerprint
6. Compare the band patterns

DNA profiles can be used in:**→ Paternity Test:**

- DNA profile of a child is compared with potential fathers' DNA profile

	- During fertilization, half of the DNA comes from each parent, so a child will share half of their DNA with a parent - The similar the more bands, the greater the genetic similarity → Forensic Science: - Regions of DNA known as short tandem repeats (STR) are examined - Short tandem repeats consist of short repeating sections and are also known as micro-satellites - The greater the number of these regions examined, the more reliable the evidence provided → Selective or Captive Breeding Programs of animals or plants: - DNA profiles of particular organisms can be compared to determine which are genetically most different from each other - Then these organisms will be crossbred, so inbreeding will not occur																																																									
6.20	understand how to determine the time of death of a mammal by examining the extent of decomposition, stage of succession, forensic entomology, body temperature and degree of muscle contraction																																																									
	<table border="1"> <caption>Approximate data points from the decomposition mass graph</caption> <thead> <tr> <th>Days</th> <th>Mass (kg)</th> <th>Stage</th> </tr> </thead> <tbody> <tr><td>0</td><td>23</td><td>Fresh</td></tr> <tr><td>4</td><td>23</td><td>Fresh</td></tr> <tr><td>5</td><td>14</td><td>Bloated</td></tr> <tr><td>6</td><td>14</td><td>Bloated</td></tr> <tr><td>7</td><td>13</td><td>Bloated</td></tr> <tr><td>8</td><td>12</td><td>Bloated</td></tr> <tr><td>9</td><td>12</td><td>Bloated</td></tr> <tr><td>10</td><td>9</td><td>Active</td></tr> <tr><td>11</td><td>8</td><td>Active</td></tr> <tr><td>12</td><td>7</td><td>Active</td></tr> <tr><td>13</td><td>7</td><td>Active</td></tr> <tr><td>14</td><td>5</td><td>Advanced</td></tr> <tr><td>15</td><td>3</td><td>Advanced</td></tr> <tr><td>16</td><td>3</td><td>Advanced</td></tr> <tr><td>17</td><td>2</td><td>Advanced</td></tr> <tr><td>18</td><td>2</td><td>Advanced</td></tr> <tr><td>19</td><td>2</td><td>Advanced</td></tr> <tr><td>20</td><td>2</td><td>Advanced</td></tr> </tbody> </table> (iii) Explain the changes in mass of this mammal during decomposition. (4) Extent of Decomposition - Very little change at the start because body not colonized by insects - Small decrease at the end of bloated stage due to wild animals eating flesh / Small volume of gas released / Decomposition by microorganisms start - Large decrease in mass at the beginning of active stage as holes in flesh allow gases to escape - Decrease in mass during active and advanced stages as insects eat the flesh - Some mass remains because bones cannot be digested → Ambient temperature affects stage of decomposition	Days	Mass (kg)	Stage	0	23	Fresh	4	23	Fresh	5	14	Bloated	6	14	Bloated	7	13	Bloated	8	12	Bloated	9	12	Bloated	10	9	Active	11	8	Active	12	7	Active	13	7	Active	14	5	Advanced	15	3	Advanced	16	3	Advanced	17	2	Advanced	18	2	Advanced	19	2	Advanced	20	2	Advanced
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- (v) The table shows data on some of the insects collected from this decomposing mammal.

Type of insect	Percentage of some of the insect found at each stage of decomposition (%)				
	Fresh	Bloated	Active	Advanced	Remains
<i>Lucilia</i>	100.0	61.4	1.8	3.7	0.0
<i>Cochliomyia</i>	0.0	15.5	68.4	35.1	0.0
<i>Chrysomya</i>	0.0	6.4	9.8	1.2	0.0
<i>Ophyra</i>	0.0	0.0	0.6	30.9	83.9
<i>Fannia</i>	0.0	0.0	0.0	0.1	0.0
Other types of insects	0.0	16.7	19.4	29.0	16.1

Explain how these results illustrate succession.

Use the information in the table to support your answer.

(3)

Stage of Succession

- Succession means that the species of insects found on the decomposing mammal will change with time
- *Lucilia* appears on the body first (Pioneer Species)
- Then *Cochliomyia* appear on the body and *Lucilia* numbers decreases as they are outcompeted
- Then *Ophyra* arrives and *Cochliomyia* decrease in number as they are outcompeted as well

Forensic Entomology

- Study of insect life cycle by collecting eggs and hatched to identify the species and determine time of hatching
- Insects colonize the body in a specific sequence
- Stage in life cycle depends on ambient temperature

More reliable than other methods as

- Body temperature is not accurate as it would have reached ambient temperature
- Extent of rigor would not be accurate as the muscles do not stay stiff for long
- Extent of decomposition is not accurate as it depends on present of oxygen, ambient temperature and numbers of decomposers that colonize the body

Body Temperature

To estimate time of death:

- Determine temperature of the liver / rectal using a thermometer
- Use calibration curve and the temperature to estimate the time after death
- Subtract this value from the time of finding the corpse
- Use in conjunction with other methods to be more reliable

Factors affecting body temperature:

- Fat level
- Mode of death
- Surface area: Volume ratio
- Clothing and covering

	Degree of Muscle Contraction - State of muscle contraction / stiffness - Occurs due to muscle cells deficient in ATP to relax so permanent contraction occurs - Changes in rigor occurs only in the first 24-48 hours after death as muscle relaxation occurs due to enzyme released by lysosomes Factors affecting rigor: - Amount of ATP stored in muscles at the time of death - Ambient temperature
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Notice

- In this booklet, mark scheme points from past paper questions (mostly 2020-2023 and a few from 2010-2019) are included for each specification point to connect the syllabus and the past paper questions.
- For less frequently examined specification points, Pearson Edexcel IAL textbook and Save My Exams (available at [savemyexams.com](https://www.savemyexams.com)) notes have been the main reference of this booklet.
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